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Course No : EEE 4603
Assignment No : 01
CEP Name : Design & Implementation of an Intelligent Automated Packing System
Date of Submission : 13/09/2026

Executive Summary

This report covers the design and testing of a PLC-based control system for the automated bottle packaging line described in the EEE 4603 CEP brief, based on Akij Food & Beverage Ltd. (AFBL)'s bottling line. The system controls four stations on a single conveyor, Quality (Fill-Level) Inspection, Bottle Capping, Bottle Counting, and Batch Packaging, and the whole thing is programmed as ladder logic in XG5000 for an LS Electric PLC.

The core of the design is a single master interlock, M00000 (System Run), that controls every actuator on the line. If Stop, Emergency Stop, or either of the two faults in the program is active, M00000 drops and the conveyor, capper, and packager are all cut off straight away. The two faults are an Underfill Fault, which stops the line and turns on an alarm when a bottle doesn't reach the required fill level, and a Packaging Timeout Fault, which I added myself to catch a stalled or jammed packaging cycle using a supervisory timer. Both faults require the operator to press a Fault Reset button before the line can restart, and Start has to be pressed again afterwards, so acknowledging a fault is never enough on its own to get the conveyor moving.

All the timing and counting values come from my student ID, 220021116. The capping cycle is fixed at 21.0 seconds (last two digits of the ID, 16, plus 5) and a batch is 16 bottles (last digit, 6, plus 10). I compared two possible fault-recovery approaches for the underfill case, an automatic reject arm and a manual stop-and-clear procedure, and went with the manual approach because it is simpler to build, safer since there are no extra moving parts near the bottles, and fits the brief's requirement that a faulty bottle be handled without damaging the bottles around it.

The finished program was tested in the XG5000 simulator for the three required scenarios, normal operation, an underfill fault and recovery, and an emergency stop mid-cycle, plus a fourth test for the packaging timeout fault I added on my own. Screenshots of each test are included in Section 9.

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1 Introduction

Akij Food & Beverage Ltd. (AFBL) currently caps and packages its bottles by hand, and this has led to a high rate of rejected product and regular bottlenecks on the line. To fix this, the plant wants to automate the line under the supervision of a central Programmable Logic Controller (PLC). The conveyor needs to move bottles continuously through four stations: Quality (Fill-Level) Inspection, Bottle Capping, Bottle Counting, and Batch Packaging.

For this assignment I took on the role of the automation engineer and had to design, program, simulate, and document a complete PLC control solution in XG5000 ladder logic. Along with the four stations, the brief also asks for safe handling of underfilled bottles, prevention of jams or collisions at the packaging station, Stop/E-Stop safety interlocks, and reliable batch counting during continuous operation. This report walks through how I approached each of these, from picking sensors through to verifying the finished program in simulation.

2 Project Objectives

The objectives I addressed for this Complex Engineering Problem are:

- Select and justify appropriate sensors for fill-level inspection, capping-station bottle presence, bottle counting, and packaging-completion confirmation.
- Design a complete I/O address allocation scheme and control architecture for the packaging line.
- Build a master control (System Run) interlock that safely governs all actuators under Start, Stop, Emergency Stop, and fault conditions.
- Implement timing and counting logic whose presets are derived from my Student ID (220021116).
- Evaluate at least two fault-recovery strategies and implement the one I select in the PLC program.
- Identify and design a solution for at least one abnormal operating condition beyond what is explicitly stated in the brief.
- Verify the complete system in the XG5000 simulator under normal operation, fault interruption/recovery, and emergency-stop conditions.

3 Sensor/Hardware Selection

Bottling and packaging lines are a well-established application for non-contact photoelectric, capacitive, and inductive sensors. I picked the sensors below based on four things that matter for this kind of line: staying non-contact so the product isn't touched, a response time that suits the conveyor speed, an enclosure that can survive a humid, washdown-prone plant floor (IP65), and a straightforward digital signal that plugs into the PLC's discrete inputs without any extra signal conditioning.

3.1 Fill-Level (Underfill) Inspection Sensor

For the fill-level check, I am using a through-beam (or retro-reflective) photoelectric sensor mounted at the height that corresponds to the minimum acceptable fill line. As a bottle passes, the beam only gets broken if the liquid reaches that height, so an underfilled bottle simply fails to break the beam and gets flagged. I picked this over capacitive or ultrasonic level sensing because it does not touch the product at all, it reacts fast enough for the conveyor speed, and it is a cheap way to get a simple pass/fail digital signal into the PLC (input P00005).

3.2 Capping-Station Bottle-Presence Sensor

Bottle presence at the capping station (P00003) is picked up with a diffuse photoelectric sensor with background suppression. I did not go with an inductive sensor here since the bottles are PET/plastic and would not reliably trigger one. Background-suppression photoelectric sensors work regardless of bottle colour, label, or the lighting on the plant floor, which is why they are the standard choice for this kind of detection.

3.3 Bottle Counting Sensor

Counting (P00006) uses a high-speed through-beam photoelectric sensor set up to give one clean pulse per bottle. Through-beam sensors give a sharper, more repeatable switching edge than diffuse ones, which matters here since a missed or doubled pulse would throw off the whole batch count.

3.4 Packaging-Completion Confirmation Sensor

To confirm the packaging cycle has finished (P00007), I am using an inductive proximity sensor or a mechanical limit switch at the end of the actuator's stroke. This spot sees a lot of mechanical wear and possible debris, so a proximity/limit-switch is a better fit than an optical sensor. A lot of commercial packaging machines already expose this as a dry-contact "cycle complete" signal, which makes it easy to wire straight into the PLC.

3.5 Hardware Selection Summary

Table 1: Sensor and hardware selection summary

Station	Parameter Sensed	Sensor Technology	PLC Input
Quality Inspection	Bottle fill level	Through-beam photoelectric (IP65)	P00005
Capping Station	Bottle presence	Diffuse photoelectric, background suppression	P00003
Counting Station	Bottle pass-through	Through-beam photoelectric (high speed)	P00006
Packaging Station	Cycle-complete confirmation	Inductive proximity / limit switch	P00007

Note: I am assuming the PLC platform (LS Electric, programmed in XG5000) and its discrete I/O modules run on standard 24 VDC sinking/sourcing logic, which matches the sensors listed above.

4 System Architecture and I/O Allocation

4.1 Control Philosophy

Everything in this program runs through one master bit, M00000 (System Run), which gates every actuator. M00000 only turns on when Start is pressed while Stop, Emergency Stop, and both fault latches are inactive, and it stays latched until any one of those conditions becomes true. Because every actuator output is gated by this same bit, there is no way for Stop, E-Stop, or a fault to get bypassed by some other part of the program. Both faults also have to be

cleared and Start pressed again before M00000 can turn back on, so simply clearing a fault is never enough on its own to restart the conveyor. Clearing a fault and authorising the line to move are kept as two separate actions on purpose.

4.2 Digital Input Allocation

Table 2: Digital input allocation

Address	Signal Name	Device Type	Logic Notes
P00000	Start Push Button	NO pushbutton	Momentary; seals M00000 in Network 1
P00001	Stop Push Button	NC pushbutton (fail-safe)	Wired normally-closed, so a broken wire is treated the same as a stop request
P00002	Emergency Stop	NC E-Stop (fail-safe)	Same fail-safe wiring as Stop
P00003	Capping Station Bottle-Present Sensor	Diffuse photoelectric (NO)	Used as a rising-edge (pulse) contact so it does not re-trigger
P00005	Fill-Level (Underfill) Sensor	Through-beam photoelectric (NO)	Only energises when an underfill is detected
P00006	Bottle Counting Sensor	Through-beam photoelectric (NO)	Rising-edge (pulse) contact, one pulse per bottle
P00007	Packing-Complete Signal	Inductive proximity / limit switch (NO)	Confirms the packaging actuator has finished its stroke
P00008	Fault Reset Push Button	NO pushbutton	Manually clears both fault latches (M00002, M00004)

Note: address P00004 is not used in the final design and is left free for future expansion, for example a second interlock sensor at the capping station.

4.3 Digital Output Allocation

Table 3: Digital output allocation

Address	Signal Name	Device Type	Energised When
P00040	Conveyor Motor	3-phase motor contactor	M00000 is on and capping/fault/packaging are not in progress
P00041	Capping Machine	Solenoid / actuator	M00000 AND Capping Active (M00001)
P00042	Alarm Lamp (stack light)	Red beacon	Underfill Fault (M00002) OR Packaging Timeout Fault (M00004)
P00043	Packaging Machine	Actuator / cylinder	M00000 AND Packaging Active (M00003)

4.4 Internal Memory, Timer, and Counter Allocation

Table 4: Internal memory, timer, and counter allocation

Address	Description	Type / Preset	Function
M00000	System Run	Latch (seal-in)	Master control interlock
M00001	Capping Active	Latch (seal-in)	Enables the capping actuator and its timer
M00002	Underfill Fault	Latch (seal-in)	Halts the system on an under-filled bottle
M00003	Packaging Active	SET/RESET latch	Enables the packaging actuator; blocks new bottles from entering
M00004	Packaging Timeout Fault	Latch (seal-in)	Halts the system if the packaging cycle stalls or jams
T0000	Capping Timer	TON, K210 (21.0 s)	Fixed capping duration
T0001	Packaging Timeout Timer	TON, K300 (30.0 s)	Watchdog on the packaging cycle
C0000	Batch Counter	CTU, K16	Counts capped bottles toward the batch size

4.5 Parameter Derivation from Student ID

As required in the brief, every timing and counting value below comes from my Student ID, 220021116.

Table 5: Parameter derivation from Student ID

Parameter	Formula	Calculation	Value Used
Capping Duration (XX)	Last 2 digits of ID + 5	16 + 5	21.0 s $\rightarrow$ T0000 = TON K210 (0.1 s base)
Batch Size (YY)	Last digit of ID + 10	6 + 10	16 bottles $\rightarrow$ C0000 = CTU K16

The Packaging Timeout Timer (T0001, 30.0 s) is not derived from the ID. It is a margin I chose myself as an engineering design decision, explained in Section 7.

5 Sequence of Operation

5.1 Normal Operating Sequence

1. The operator presses Start (P00000). As long as Stop, Emergency Stop, and both faults are inactive, M00000 (System Run) turns on and latches.
2. The conveyor (P00040) starts running and bottles move toward the inspection and capping stations.
3. When a bottle reaches the capping-station sensor (P00003), Capping Active (M00001) latches on the rising edge, the conveyor stops, and the capping actuator (P00041) runs for the fixed 21.0 s capping time (T0000).

4. Once the timer finishes, M00001 resets, the capper turns off, and the conveyor starts moving again on its own.
5. Every capped bottle that passes the counting sensor (P00006) adds one to the batch counter (C0000).
6. Once the counter reaches 16, Packaging Active (M00003) turns on, the conveyor stops, and because M00003 also blocks the conveyor output, no new bottle can reach the packaging station while it is running, which is what prevents a collision there.
7. The packaging machine (P00043) keeps running until it gives the Packing-Complete signal (P00007). At that point M00003 and the batch counter both reset and the conveyor starts up again for the next batch.
8. This cycle (steps 2–7) just keeps repeating until Stop/E-Stop is pressed or a fault is triggered.

5.2 Fault and Interrupt Sequences

Underfill Fault: if the fill-level sensor (P00005) picks up an underfilled bottle, M00002 latches right away, the alarm (P00042) turns on, and M00000 drops, cutting power to the conveyor, capper, and packager instantly, no matter what station was running at the time. The line stays down until someone removes the bad bottle, presses Fault Reset (P00008) to clear M00002, and presses Start again.

Packaging Timeout Fault: if Packaging Active (M00003) stays on for longer than the 30.0 s watchdog (T0001) without a Packing-Complete signal, M00004 latches, the alarm turns on, M00000 drops, and M00003 is force-reset once the operator presses Fault Reset, so the batch can be re-attempted cleanly once the jam is cleared.

Emergency Stop / Stop: hitting either button drops M00000 straight away through Network 1. Since every actuator output is gated by M00000, the conveyor, capper, and packager all lose power in the same PLC scan, whether or not a capping or packaging cycle was running at the time.

6 Fault-Recovery Strategy Selection and Justification

I looked at two ways of handling an underfilled bottle at the Quality Inspection station.

6.1 Strategy A — Automatic Inline Reject Arm

A pneumatic or servo-actuated arm automatically pushes a detected underfilled bottle off the conveyor into a reject bin, so the line keeps running without an operator having to step in.

6.2 Strategy B — Manual Operator Halt-and-Clear

The conveyor stops completely and an alarm goes off the instant an underfilled bottle is detected. An operator removes the bottle by hand, presses Fault Reset, and presses Start to get the line moving again.

6.3 Comparative Evaluation

Table 6: Comparative evaluation of fault-recovery strategies

Criterion	A: Automatic Reject Arm	B: Manual Halt-and-Clear
Safety	Adds a moving actuator near bottles and operators, so it needs its own guarding and risk assessment	No extra moving parts; the line is fully stopped before anyone touches it
Downtime Cost	Very low, the line keeps running	Higher, the whole line stops every time
Implementation Complexity	High, needs a reject actuator, its own sensing/timing, and extra PLC logic	Low, built entirely from the I/O already in the design plus one latch/reset pair
Throughput Impact	Small, as long as rejects are infrequent	Scales with how often the fault happens, acceptable if it stays rare
Risk of Secondary Faults	The reject arm itself could jam or misfire and damage a nearby bottle	None, since no extra actuator is added

6.4 Selected Strategy and Justification

I went with Strategy B, the manual halt-and-clear approach, and built it into the program as the Underfill Fault latch, M00002, in Section 8. A few reasons for this. The brief specifically asks for a faulty bottle to be handled without damaging the bottles around it, and there is no risk of that with a manual removal since nothing else on the line is moving at the time. It is also a much simpler build for the scope of this CEP, since it only needs the I/O already listed and no additional actuator hardware. And stopping the whole line before anyone touches it is just safer than having an operator reach near a still-moving conveyor or an active reject mechanism. The trade-off is that the line stops completely every time this happens, but since underfill faults should be rare if the filling stage upstream is working correctly, that seemed like a reasonable trade to make.

7 Additional Abnormal Operating Condition: Packaging-Station Timeout / Jam

7.1 Description and Risk

Beyond the underfill case that is explicitly in the brief, I think the packaging station itself is a likely place for something to go wrong, either the actuator jams mechanically or the Packing-Complete sensor (P00007) fails to trigger. In the base design, Packaging Active (M00003) would just stay latched forever in either case. That halts the conveyor, since M00003 blocks it, but gives the operator no real indication of why, which is a silent failure that could go unnoticed for a while.

7.2 Detection Strategy

To catch this, I added a watchdog timer, T0001, that runs for as long as M00003 is on. If the Packing-Complete signal does not arrive before the timer hits its preset, a separate fault bit, M00004, latches. I set the preset to 30.0 seconds, comfortably above how long a normal packaging cycle should take, so it will not trip during ordinary cycle-time variation but will still catch a genuine stall quickly enough to limit lost production and wear on the actuator.

7.3 Recovery Strategy

Once M00004 latches, it turns on the shared alarm (P00042) and drops M00000, the same way the underfill fault does, so the conveyor stops and the packaging actuator is cut. The operator clears the physical jam, presses Fault Reset (P00008), which clears M00004 and also force-resets M00003 (Network 12), and the line is then ready to restart with Start.

7.4 Justification

I chose this over something like a mechanical shear-pin or a separate thermal overload relay because it does not need any extra hardware beyond a timer instruction the PLC already supports, it gives a clear, visible fault instead of a silent stall, and it works the same way as the underfill fault, sharing the same reset button and alarm, so the operator does not have to learn a second way of clearing faults. Catching a jam within 30 seconds instead of relying on someone noticing the line has gone quiet should also help cut down on wasted time and unnecessary wear on the actuator.

8 PLC Ladder Logic Design (XG5000)

8.1 Design Conventions

A few conventions apply throughout the program:

- Stop (P00001) and Emergency Stop (P00002) are both wired normally-closed, so the input is normally ON and only goes OFF when the button is pressed or the wire is cut. That way a broken wire is treated the same as someone hitting Stop.
- All timers use a 0.1-second time base, so a preset of K210 corresponds to 21.0 s.
- Rising-edge (pulse) contacts, marked |P|, are used anywhere a single bottle passing a sensor should trigger just one action, so it does not keep re-triggering while the bottle is still sitting at the sensor.
- Every actuator, the conveyor, capper, and packager, is gated directly by M00000, so Stop, E-Stop, or either fault cuts power to all three at once.

The full program is shown below as fifteen networks, grouped by function, with the corresponding XG5000 screenshots underneath each group.

8.2 Networks 1–3: Master Control and Capping Sequence

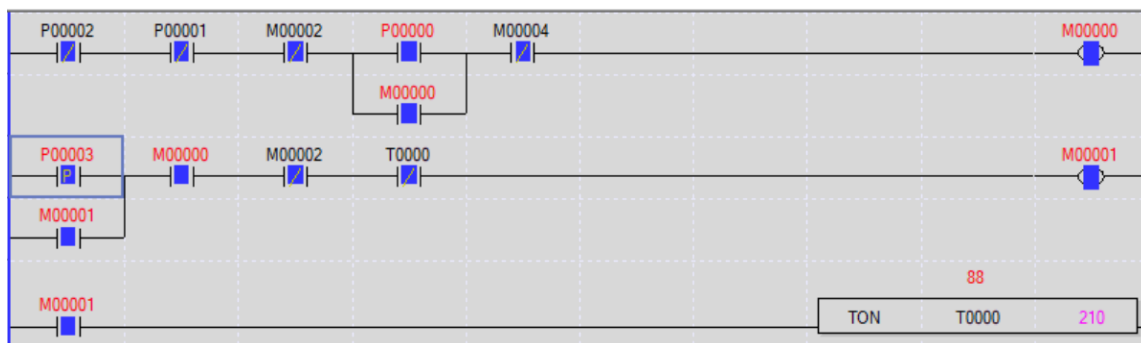


Figure 1: XG5000 ladder logic — System Run, Capping Active, and Capping Timer.

8.3 Networks 4–6: Underfill Fault, Alarm, and Counting

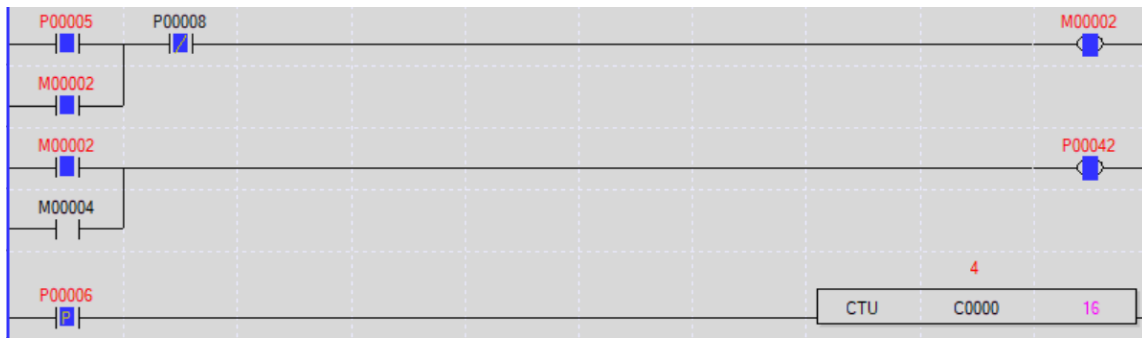


Figure 2: XG5000 ladder logic — Underfill Fault Latch, Alarm Lamp, and Bottle Counter.

8.4 Networks 7–9: Batch Packaging Control

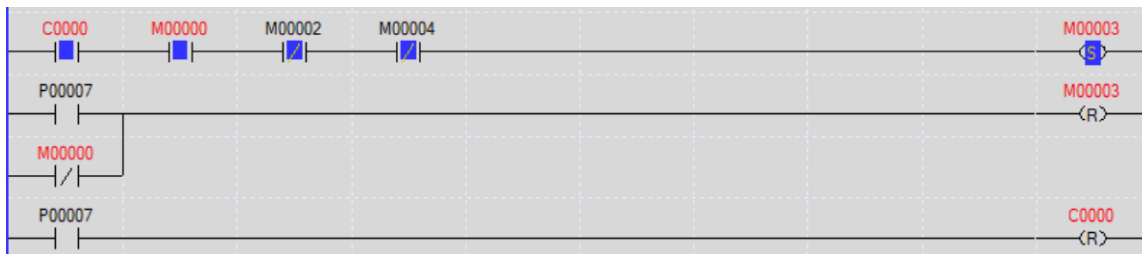


Figure 3: XG5000 ladder logic — Start Packaging, Packing Complete, and Counter Reset.

8.5 Networks 10–12: Packaging Timeout Fault (Additional Condition)

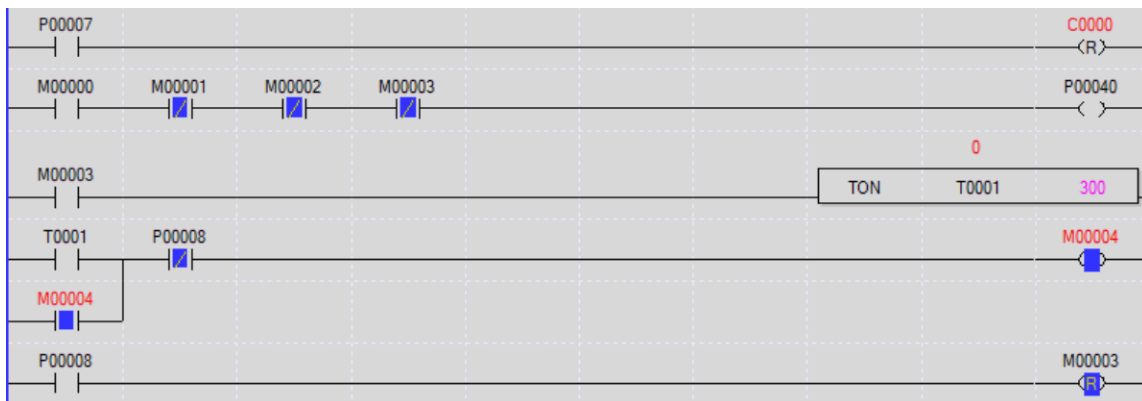


Figure 4: XG5000 ladder logic — Packaging Timeout supervisory fault subsystem.

8.6 Networks 13–15: Output Interlocking

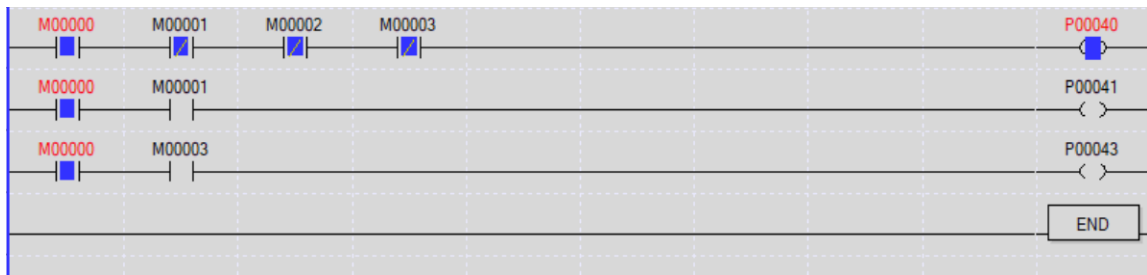


Figure 5: XG5000 ladder logic — final actuator output interlocking.

9 Simulation and Verification

I ran the finished program through the XG5000 simulator for the three scenarios the brief asks for, plus a fourth one for the packaging-timeout fault I added on my own. For each test, I have described what should happen first, followed by the screenshots from that run.

9.1 Test Case 1 — Normal Continuous Operation

What should happen: Start latches M00000 and gets the conveyor moving. A bottle at the capping sensor starts the 21.0 s capping cycle, after which the conveyor starts again on its own. The counter goes up with every capped bottle, and once it reaches 16, packaging starts and the conveyor stops until Packing-Complete comes in, at which point the counter resets and the whole thing repeats.

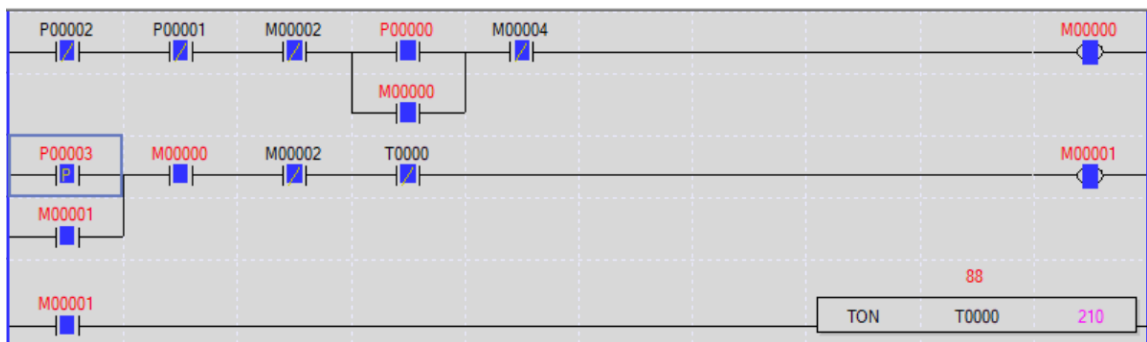


Figure 6: Simulation — normal continuous operation, capping cycle in progress.

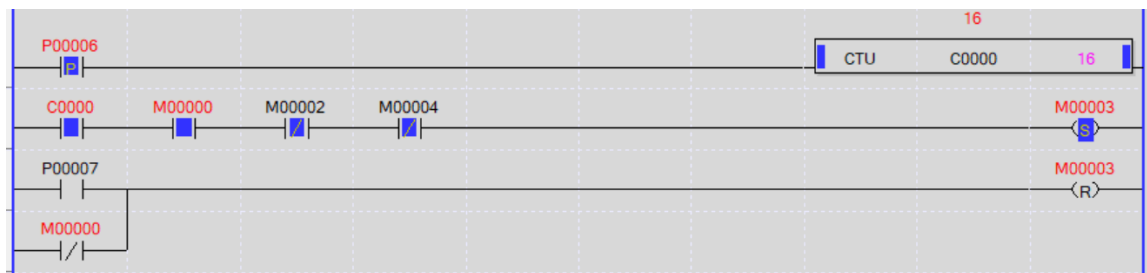


Figure 7: Simulation — batch packaging cycle triggered at 16 bottles.

9.2 Test Case 2 — Underfill Fault Interruption and Recovery

What should happen: forcing P00005 ON sets M00002 right away, drops M00000, and cuts power to the conveyor, capper, and packager while turning on the alarm (P00042), no matter what was running at the time. It stays down until P00008 clears M00002 and Start is pressed again.

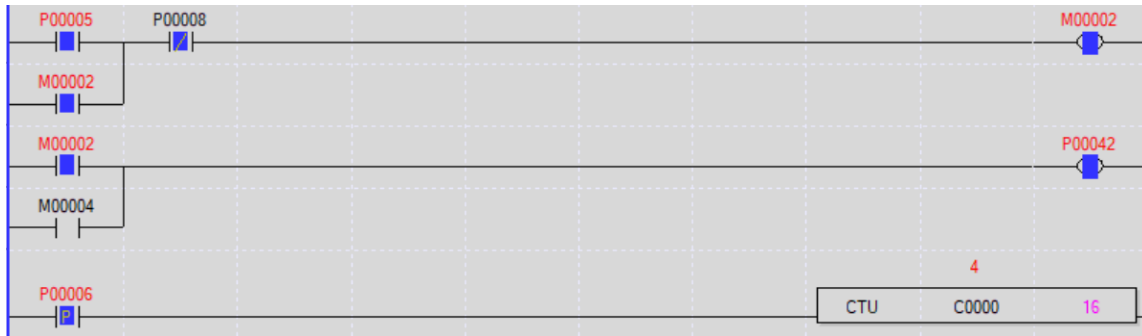


Figure 8: Simulation — underfill fault latched, line halted, alarm lamp active.

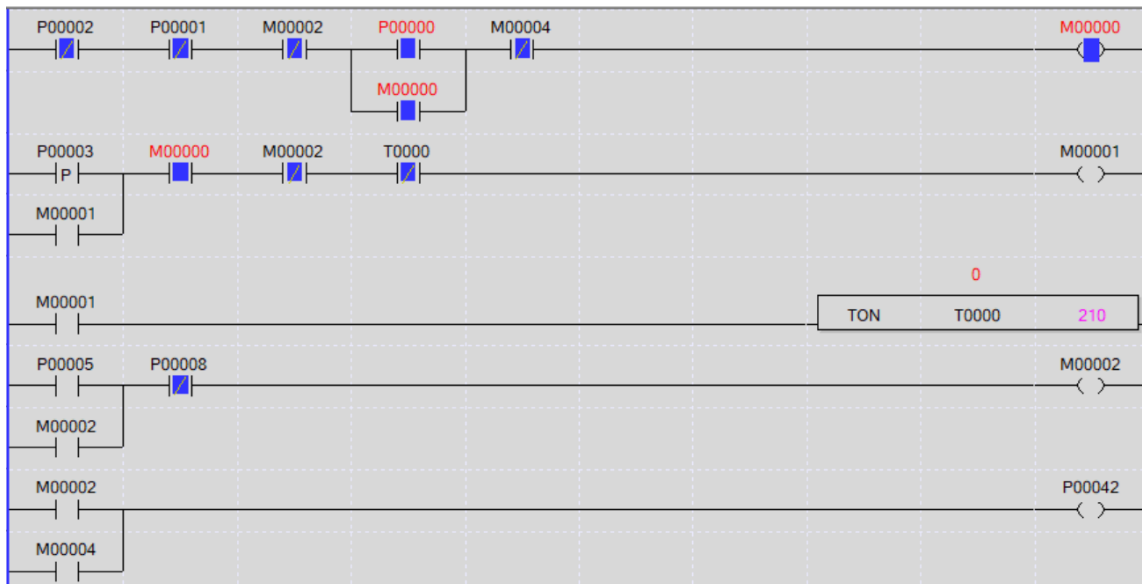


Figure 9: Simulation — underfill fault cleared and system successfully restarted.

9.3 Test Case 3 — Emergency Stop During an Active Capping/Packing Cycle

What should happen: forcing P00002 OFF while M00001 (capping) or M00003 (packaging) is active drops M00000 straight away, which cuts the conveyor, capper, and packager in the same scan. The timers reset once M00001/M00003 turn off.

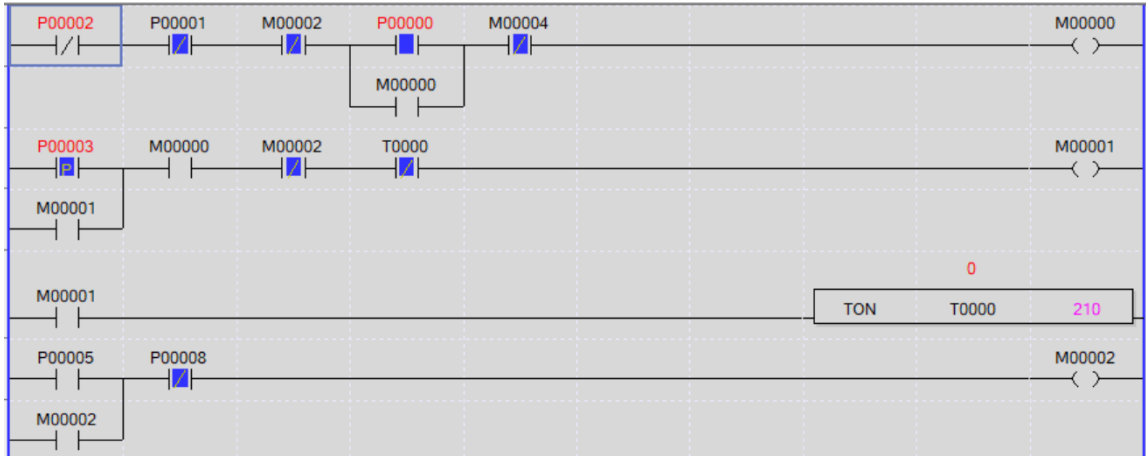


Figure 10: Simulation — emergency stop interrupting an active capping cycle.

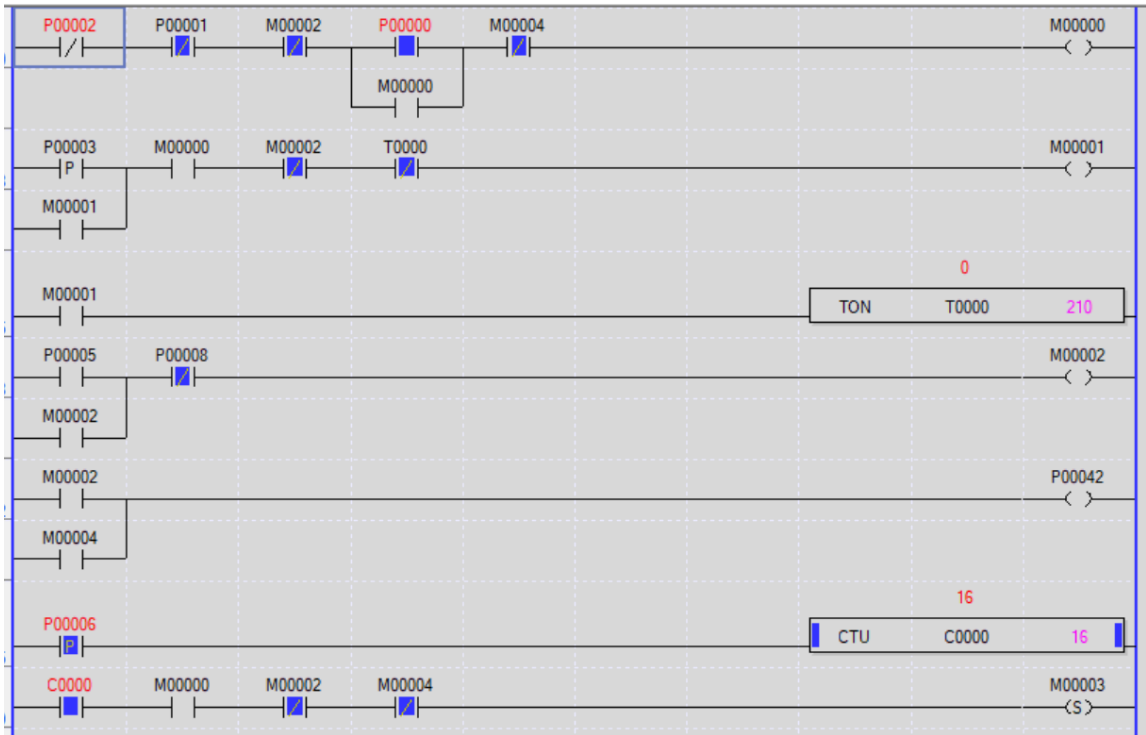


Figure 11: Simulation — emergency stop interrupting an active packaging cycle.

9.4 Test Case 4 — Packaging-Station Timeout Fault (Additional Condition)

What should happen: with M00003 on and P00007 held off past 30.0 s, T0001 hits its preset, M00004 latches, the alarm turns on, and M00000 drops. Pressing P00008 clears M00004 and also force-resets M00003 through Network 12.

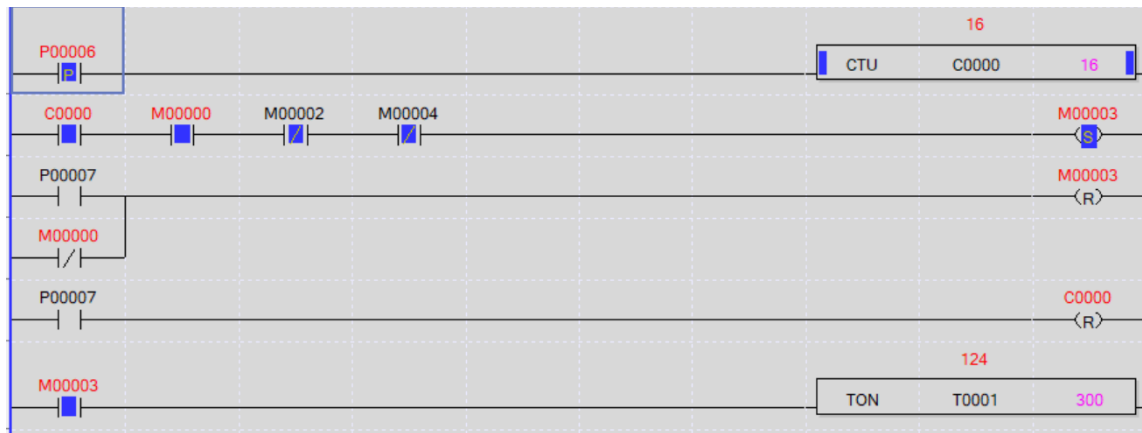


Figure 12: Simulation — packaging-station timeout fault detected and latched.

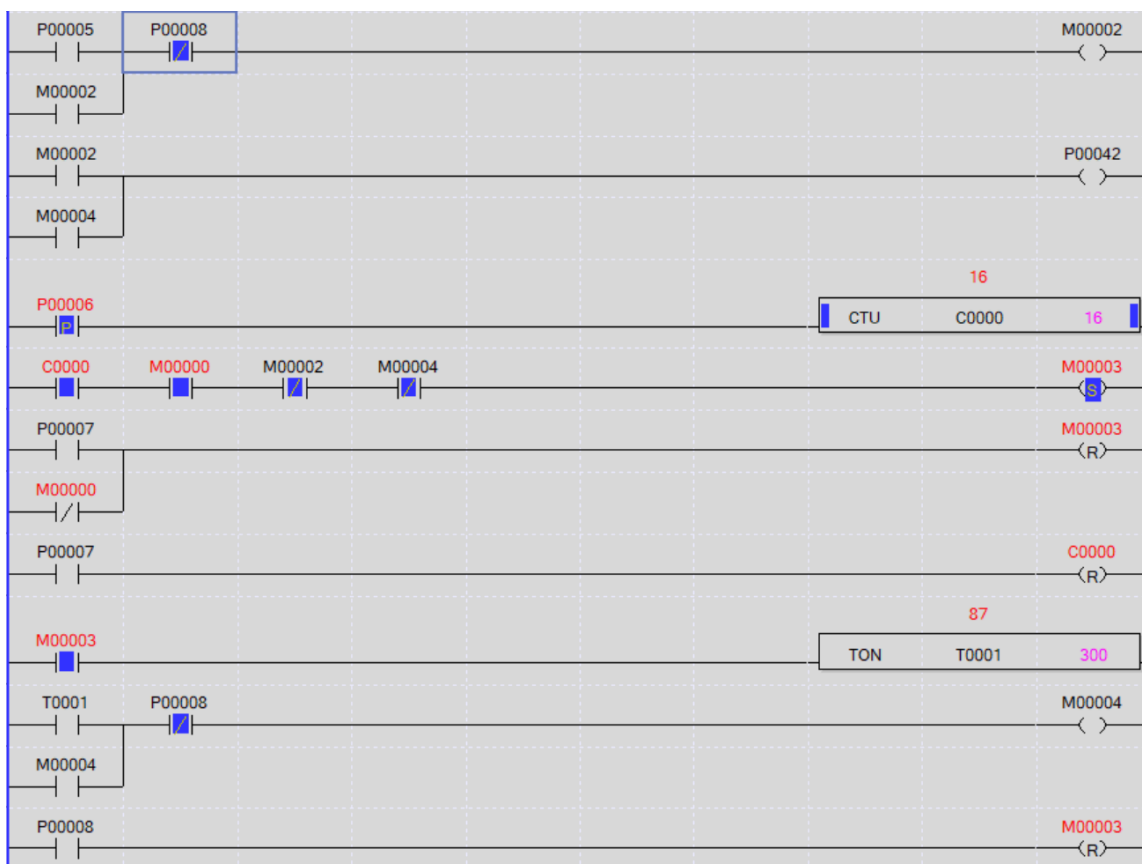


Figure 13: Simulation — packaging timeout fault cleared, system ready to restart.

10 Discussion and Limitations

This design covers everything the brief asks for using just discrete I/O and standard ladder instructions, contacts, coils, SET/RESET, TON, and CTU, without needing any analogue modules, an HMI, or communication protocols. That keeps it cheap enough to run on a small PLC for a single packaging line.

There are a few limitations worth mentioning. Everything here was only tested in the XG5000 simulator, so real sensor debounce, electrical noise, and actual mechanical response times were

not part of this and would still need to be checked once it is actually commissioned. The two fault types also share one alarm lamp, so an operator cannot tell which fault happened just by looking at it, a second lamp or a basic HMI screen would fix that. The manual halt-and-clear approach also trades some throughput for safety and simplicity, if underfill faults turn out to happen a lot more often than expected, switching to an automatic reject arm (Section 6.1) would probably make more sense in production, and since each fault is just its own latch feeding into the shared alarm and the master interlock, that change would not need much of the program to be rebuilt.

11 Conclusion

This report has gone through the full design of a PLC control system for AFBL's packaging line: sensor choices, I/O allocation, the control architecture, the fault-recovery comparison, and the complete XG5000 ladder logic. Every timing and counting value comes from my Student ID (220021116), giving a 21.0 s capping cycle and a 16-bottle batch. I implemented the manual halt-and-clear strategy along with an extra packaging-timeout watchdog of my own design, and tested all of it in simulation across normal operation, fault recovery, and emergency-stop scenarios. I believe this gives a solid, safe automated solution for the problem set out in the brief.

A Complete Ladder Logic Listing (Consolidated)

The image below shows the complete, consolidated ladder logic program, all fifteen networks, as built in XG5000, included here alongside the separate .GPJ project file.

